**Question 7**

| Question | Answers | Extra information | Mark | AO / Spec Ref. |
|---|---|---|---|---|
| **07.1** | arteries | | 1 | AO1<br>4.2.2.2<br>4.2.2.4 |

| Question | Answers | Extra information | Mark | AO / Spec Ref. |
|---|---|---|---|---|
| **07.2** | pushes / moves blood<br><br>to get oxygen around the body | <br><br>allow description of getting oxygen around the body, such as through blood vessels<br>**or**<br>to get oxygen to a named organ<br>do **not** accept to get oxygen to the lungs<br><br>ignore reference to restarting the heart<br>ignore reference to the pacemaker | 1<br><br>1 | AO2<br>4.2.2.2<br>4.4.2.1 |

| Question | Answers | Extra information | Mark | AO / Spec Ref. |
|---|---|---|---|---|
| **07.3** | provides oxygen (for respiration) | allow idea of carbon dioxide triggering breathing to restart | 1 | AO2<br>4.2.2.2<br>4.4.2.1 |

| Question | Answers | Extra information | Mark | AO / Spec Ref. |
|---|---|---|---|---|
| **07.4** | statin(s) | allow named statin | 1 | AO1<br>4.2.2.4 |

| Question | Answers | Extra information | Mark | AO / Spec Ref. |
|---|---|---|---|---|
| **07.5** | (stent) opens / widens (blocked blood) vessel | allow (stent) keeps (blocked blood) vessel open<br>allow a description of the blood vessel being opened<br>ignore type of blood vessel<br>ignore unblocks (blood) vessel | 1 | AO1<br>4.2.2.4 |
| | to allow (more) blood to flow<br>**or**<br>to allow (more) glucose / oxygen to the heart (cells / tissue / muscle) | | 1 | |

| Question | Answers | Extra information | Mark | AO / Spec Ref. |
|---|---|---|---|---|
| **07.6** | any **two** from:<br>• smoking increases the (%) risk of **all** types of (cardiovascular) disease<br>• smoking increases the (%) risk of having (disease) **H** more than any other type of (cardiovascular) disease<br>• smoking increases the (%) risk of having (disease) **E** less than any other type of (cardiovascular) disease | ignore smoking causes (cardiovascular) disease<br>allow not smoking decreases the risk of **all** types of (cardiovascular) disease<br>allow if you smoke, you are **most** likely to get (disease) **H**<br>allow if you smoke, you are **least** likely to get (disease) **E**<br>allow a comparison of the effect of smoking on the risk of two (cardiovascular) diseases<br>allow two comparisons of the effect of smoking on the risk of two (cardiovascular) diseases for **2** marks | 2 | AO3<br>4.2.2.6 |

| Question | Answers | Extra information | Mark | AO / Spec Ref. |
|---|---|---|---|---|
| **07.7** | *y*-axis labelled **'**Percentage / % increase in risk (compared to people who have never smoked)**'** | | 1 | AO2 4.2.2.6 |
| | correct scale of 1 cm = 5% on *y*-axis | | 1 | |
| | all bars plotted correctly | allow a tolerance of ±½ small square<br>ignore bars touching<br>ignore width of bars | 1 | |
| | all bars correctly labelled | | 1 | |

| Question | Answers | Extra information | Mark | AO / Spec Ref. |
|---|---|---|---|---|
| **07.8** | any **one** from:<br>• poor diet | ignore obesity<br>allow descriptions of poor diet eg diet high in (saturated) fat / cholesterol<br>ignore diet unqualified | 1 | AO1 4.2.2.6 |
| | • lack of exercise | allow descriptions of lack of exercise<br>allow high alcohol intake<br>allow other correct lifestyle factors such as having a stressful job | | |

| | |
|---|---|
| **Total Question 7** | **14** |

**Question 8**

| Question | Answers | Extra information | Mark | AO / Spec Ref. |
|---|---|---|---|---|
| **08.1** | nucleus | allow chromosome<br>ignore in the DNA | 1 | AO1<br>4.1.1.2<br>4.1.2.1 |

| Question | Answers | Extra information | Mark | AO / Spec Ref. |
|---|---|---|---|---|
| **08.2** | **A**, **D** and **E** | | 1 | AO1<br>4.2.2.1<br>4.2.2.2 |

| Question | Answers | Mark | AO / Spec Ref. |
|---|---|---|---|
| **08.3** | **Level 3:** Relevant points (reasons / causes) are identified, given in detail and logically linked to form a clear account. | 5–6 | AO2 |
| | **Level 2:** Relevant points (reasons / causes) are identified, and there are attempts at logical linking. The resulting account is not fully clear. | 3–4 | AO2 |
| | **Level 1:** Points are identified and stated simply, but their relevance is not clear and there is no attempt at logical linking. | 1–2 | AO1 |
| | **No relevant content.** | 0 | 4.2.2.1<br>4.4.2.1<br>4.4.2.3 |
| | **Indicative content:**<br><br>**Difficulty digesting food**<br>• less / no lipase<br>• (so) less / no fat broken down<br>  ○ into fatty acids<br>  ○ into glycerol<br><br>• less / no carbohydrase / amylase<br>• (so) less / no carbohydrate / starch broken down<br>  ○ into glucose / sugar<br><br>• less / no protease<br>• (so) less / no protein broken down<br>  ○ into amino acids<br><br>**Difficulty gaining body mass**<br>• less / no absorption<br>  ○ of small / soluble molecules<br>  ○ of fatty acids<br>  ○ of glycerol<br>  ○ of glucose / sugar<br>  ○ of amino acids<br><br>• fewer molecules **or** fewer amino acids available for building protein / muscle / cells / tissues<br>• less fat stored<br>• less respiration<br>• less energy<br>• (so less energy) for building new molecules / cells / tissues<br><br>For **Level 3** details of difficulty digesting food **and** difficulty gaining body mass are needed. | | |

| Question | Answers | Extra information | Mark | AO / Spec Ref. |
|---|---|---|---|---|
| **08.4** | large surface / area | allow large surface / area to volume (ratio) | 1 | AO1<br>4.1.3.1<br>4.2.2.2 |
| | (large) capillary network<br>**or**<br>good / efficient blood supply | allow many capillaries | 1 | |
| | walls are thin<br>**or**<br>walls are one cell thick | ignore references to membranes<br>ignore alveoli are thin<br>ignore alveoli are one cell thick<br>do **not** accept thin cell walls<br><br>ignore references to alveoli being moist<br>ignore steep concentration gradient | 1 | |